

ZOMATO DELIVERY OPERATIONS ANALYTICS

Business Analyst Portfolio Project 2 — Supporting Documents

Root Cause Analysis & KPI Framework | Executive Summary

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Status: Final v1.0

SECTION 1

Root Cause Analysis & KPI Framework

A structured RCA connecting patterns across five business goals to the root causes of delivery delay

ROOT CAUSE ANALYSIS & KPI FRAMEWORK

Zomato Delivery Operations Analytics

Document Title	Root Cause Analysis & KPI Framework — Zomato Delivery Operations Analytics
Project	Business Analyst Portfolio Project 2
Version	1.0
Author	Taruna Guliya, Business Analyst
Tools Used	MySQL, Power BI
Status	Final v1.0

1. Purpose & Scope

This document contains findings from a structured SQL-based analysis of Zomato's delivery operations dataset (45,584 orders) across five business goals: delivery time reduction, rider performance, resource allocation, customer experience, and operational bottlenecks. Rather than treating each goal's findings in isolation, this RCA connects patterns across goals to identify the small number of root causes that actually explain the majority of delivery delays and inconsistency.

Scope note: this dataset covers delivery operations only (Feb–Apr 2022). It contains no customer identifier or order value, so “customer experience” findings are built on an SLA-based proxy (orders exceeding 40 minutes), not direct satisfaction data. This is a structural boundary of the available data, not a data quality defect.

2. KPI Framework

The analysis is organized around one primary KPI and its four key drivers, reflecting how the underlying causes actually surfaced in the data:

- Primary KPI: On-Time Delivery Rate (orders ≤ 40 minutes)
 - Driver 1: External conditions — traffic density, weather, festival days
 - Driver 2: Rider workload — number of simultaneous deliveries assigned
 - Driver 3: Fleet condition — vehicle condition rating
 - Driver 4: Data integrity — timestamp reliability for preparation-time metrics

Supporting KPIs tracked alongside the primary metric:

KPI	Value	Why it matters
Average Delivery Time	26.29 min	Headline operational metric

KPI	Value	Why it matters
SLA Breach Rate (>40 min)	8.86%	Proxy for poor customer experience
Jam + Festival Breach Rate	92.06%	Worst-case combination scenario
Order Volume Imbalance (city)	~4.3x	Demand concentration in metros
Multi-delivery Time Penalty (3 vs 0)	+24.9 min	Largest controllable operational factor

3. Root Cause Analysis

Findings are organized from strongest/most actionable to most exploratory, and each is traced back to the specific goal(s) that produced the supporting evidence.

3.1 Root Cause: Combination of external conditions (festival + traffic)

Evidence: Goals 1 & 4

- **What we found:** Traffic density alone adds up to ~10 minutes (Jam vs. Low), and festival days alone add ~75% to delivery time (45.5 vs. 26.0 min). But these factors do not act independently — the SLA breach rate for Jam traffic on a normal day is 16.71%, while Jam traffic on a festival day jumps to 92.06%. Even Low traffic on a festival day (78% breach rate) is worse than Jam traffic on a normal day.
- **Why this is the root cause, not just a symptom:** Festival status overrides traffic as the dominant driver of poor delivery experience. This is a combining effect, not two separate problems — meaning a fix aimed only at traffic (e.g., route optimization) will not meaningfully help on festival days, and resourcing decisions need to specifically anticipate festival-day demand rather than scaling linearly with normal traffic patterns.

3.2 Root Cause: Rider workload (multiple simultaneous deliveries)

Evidence: Goal 5

- **What we found:** Average delivery time increases steadily and substantially with delivery batching: 22.88 min (single delivery) → 26.85 min (+1) → 40.45 min (+2) → 47.82 min (+3). This is the single largest average-time swing of any factor tested across all five goals.
- **Why this is the root cause, not just a symptom:** This is an internal operational decision — how many deliveries a rider is assigned at once — not an external condition the business can't control. This makes it the most directly actionable lever available: adjusting delivery-batching policy has a clearer, more controllable path to improvement than trying to influence traffic or weather.

3.3 Root Cause: Combination of operational inefficiencies

Evidence: Goal 5

- **What we found:** Vehicle condition alone has a smaller but real effect (condition 0 riders average 30.07 min vs. ~24–26 min for better-maintained vehicles). When poor vehicle condition combines with multiple simultaneous deliveries, average delivery time reaches 48.42 minutes — the worst combination recorded in the entire analysis.
- **Why this matters:** This confirms the same combining pattern seen in 3.1 (festival + traffic) shows up again with a completely different pair of factors (fleet + workload). This is a pattern worth stating explicitly in the executive summary: delays in this operation are rarely caused by one factor alone — they emerge when two inefficiencies land on the same order.

3.4 Corrected finding: City-level rider inconsistency is a statistical artifact, not a real service gap

Evidence: Goals 2 & 3 (initial finding revised)

- **Initial finding (Goal 2):** Smaller cities showed a much wider gap between their best and worst rider (~14 min) compared to large metros (~4.7–5.6 min), suggesting inconsistent rider quality in smaller markets.
- **Revised after Goal 3 evidence:** Average delivery time is remarkably flat (26.0–27.0 min) across every city regardless of order volume per rider — including cities with the lowest per-rider order counts. Since resourcing/order-volume differences don't actually move the average, the wide performance gap seen in smaller cities is better explained by small sample sizes (10–16 orders per rider in small cities vs. 40–67 in metros) producing naturally noisier averages, not a genuine rider skill or training gap.
- **Why this revision matters:** This prevents a wrong recommendation — investing in rider retraining programs in smaller cities would likely have little effect, since the underlying issue is statistical data, not skill. This is flagged here specifically so it is not mistakenly acted on.

3.5 Exploratory finding: Non-obvious weather effect

Evidence: Goal 1

- **What we found:** Fog conditions (28.91 min average) are associated with slower deliveries than Stormy (25.87 min) or Sandstorm (25.88 min) conditions — the opposite of what intuition would suggest.
- **Status:** Flagged as a genuine, non-obvious pattern worth further investigation (e.g., visibility-driven speed reduction), but not built into a firm recommendation without additional data — the mechanism is plausible but unconfirmed.

3.6 Data quality limitation: Preparation-time fields are unreliable

Evidence: Goal 5

- **What we found:** Calculated preparation time (time between order placed and order picked up) produced negative values under some conditions and extreme outliers up to 1,435 minutes — both logically impossible for a food delivery order.
- **Why this is reported as a limitation, not a finding:** These values indicate timestamp inconsistencies in the source data rather than genuine restaurant performance issues. Reporting this transparently prevents a false conclusion (e.g., “certain restaurants are extremely slow”) from being drawn from unreliable fields, and flags this specific data as unsuitable for operational decisions without further cleaning.

4. Prioritized Root Causes

Ranked by combination of business impact and actionability:

Rank	Root Cause	Impact	Actionability
1	Rider workload (multi-delivery batching)	Largest single time penalty (+24.9 min)	High — internal policy lever
2	Festival + traffic compounding	92% breach rate in worst case	Medium — requires proactive staffing
3	Compounding fleet + workload issues	Worst recorded case (48.4 min)	Medium — fleet maintenance + batching policy
4	Preparation-time data integrity	Blocks a whole analysis area	Low (near-term) — needs data cleanup first

5. What This Means for Recommendations

The strongest, most defensible pattern across this entire analysis is that delivery delays in this dataset are rarely caused by a single factor — they emerge when two conditions compound (festival + traffic, or workload + fleet condition). This has a direct implication for recommendations: interventions aimed at a single factor in isolation (e.g., only optimizing routes for traffic) are unlikely to meaningfully move the on-time delivery rate. The highest-leverage actions are the ones that address compounding scenarios directly — particularly rider workload policy, since it is both the single largest factor and the one most within the business's direct control.

Detailed recommendations, expected impact, and next steps are developed in the accompanying executive summary below.

SECTION 2

Executive Summary

Recruiter-facing summary of key findings, KPIs, and recommendations

EXECUTIVE SUMMARY

Zomato Delivery Operations Analytics: Performance & Resource Optimization

Main Business Problem

Zomato seeks to optimize its delivery operations by identifying the key factors affecting delivery time, improving rider productivity, and enhancing resource allocation to deliver a faster and more reliable customer experience.

Goals

- 1. Reduce Average Delivery Time
- 2. Optimize Rider Performance
- 3. Optimize Resource Allocation Across Cities
- 4. Improve Customer Experience
- 5. Understand Operational Bottlenecks

Tool Used

MySQL, Power BI

Dataset Source

Food delivery operational dataset obtained through Kaggle containing order records, rider details, city information, traffic and weather conditions, and delivery timings used for delivery performance and operational efficiency analysis.

Dataset Contains

Columns	22
Rows	45,585
Tables	3

Data Quality Issues

- 6. **Missing values:** Rider age, rider rating, and order timestamp were missing in roughly 4–13% of records across different fields.
- 7. **Invalid ratings and ages:** Rider ratings exceeded the valid 1–5 scale (up to 6.0), and rider ages were below typical working age (as low as 15) — both indicative of data entry or extraction errors in the source.
- 8. **Scope limitation:** The source dataset does not include customer identifiers or order value, which limited this analysis to delivery operations and rider performance rather than customer retention or revenue impact.

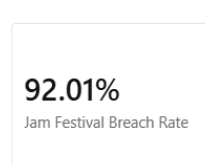
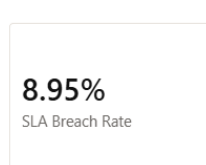
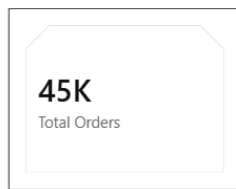
Key KPIs

9. **On-time delivery performance:** 91.14% of orders delivered within 40 minutes; SLA breaches affected 8.86% of orders.
10. **Rider workload (batching) impact:** Average delivery time rose from 22.88 min (0 deliveries) to 47.82 min (3 deliveries); the 5.1% of orders with 2+ simultaneous deliveries saw breach rates of 45.7%–100%, vs. just 4.5%–7.4% for single-delivery orders.
11. **Festival + traffic compounding:** Jam-traffic orders on festival days breached SLA 92.06% of the time, vs. 16.71% on normal days — a 5.5x jump from the same traffic condition alone.
12. **Fleet + workload compounding:** Orders with poor vehicle condition (0) and 2+ batched deliveries averaged 43.34 min, 65% slower than the 26.29 min baseline.
13. **Order volume concentration:** Metropolitan cities handled 74.8% of order volume (34,087 of 45,584 orders), vs. 22.2% urban and under 1% semi-urban — a ~4.3x metro-to-urban imbalance.
14. **City-level rider consistency:** Average delivery time stayed flat at 26.0–27.0 min across every city tier regardless of orders-per-rider volume, indicating the earlier-observed rider performance gap in smaller cities is a small-sample artifact, not a real skill gap.

Recommendations

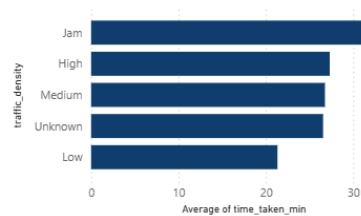
15. Cap batching at one delivery per rider — orders with 2+ simultaneous deliveries breach SLA at 45–100%, versus under 7.4% for single deliveries, despite being only 5% of volume.
16. Build a festival-specific staffing plan instead of scaling off traffic alone — Jam traffic on festival days breaches SLA 92% of the time, over 5x the same traffic on a normal day.
17. Restrict multi-delivery assignments for poor-condition vehicles — this combination raises average delivery time 65% above baseline.
18. Hold off on rider retraining in smaller cities — delivery time stays flat at 26–27 minutes across all city tiers once volume is accounted for, pointing to a sample-size effect, not a skill gap.
19. Prioritize rider capacity in metro hubs over new market expansion — metros already handle nearly 75% of order volume versus 22% urban and under 1% semi-urban.
20. Fix preparation-time timestamp errors before drawing restaurant-performance conclusions — current values include negatives and outliers past 1,400 minutes.

Dashboard Screenshots

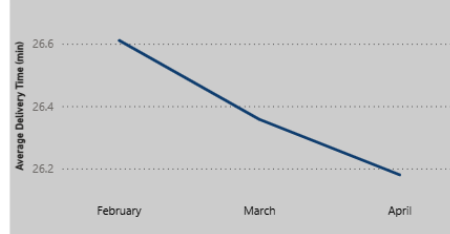


- city_name
- ☐ Agra
 - ☐ Allahabad
 - ☐ Aurangabad
 - ☐ Bangalore
 - ☐ Bhopal
 - ☐ Chennai
 - ☐ Coimbatore
 - ☐ Dehradun (inferred)

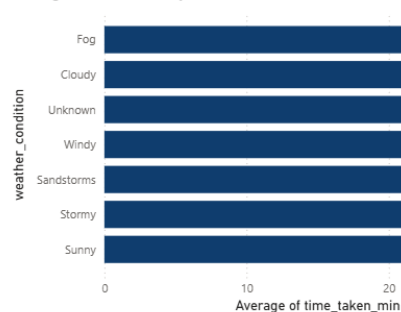
Average of time taken by traffic density



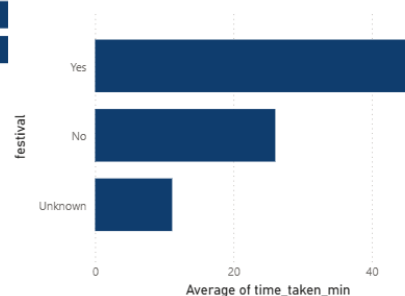
Delivery Time Trend



Average of time taken by weather condition



Average of time taken by festival



- city_name
- ☐ Agra
 - ☐ Allahabad
 - ☐ Aurangabad
 - ☐ Bangalore
 - ☐ Bhopal
 - ☐ Chennai
 - ☐ Coimbatore
 - ☐ Dehradun (inferred)

SLA Breach Rate by traffic_density and festival

